

Pask: Providing Answer before AsKing toward Proactive AI agent

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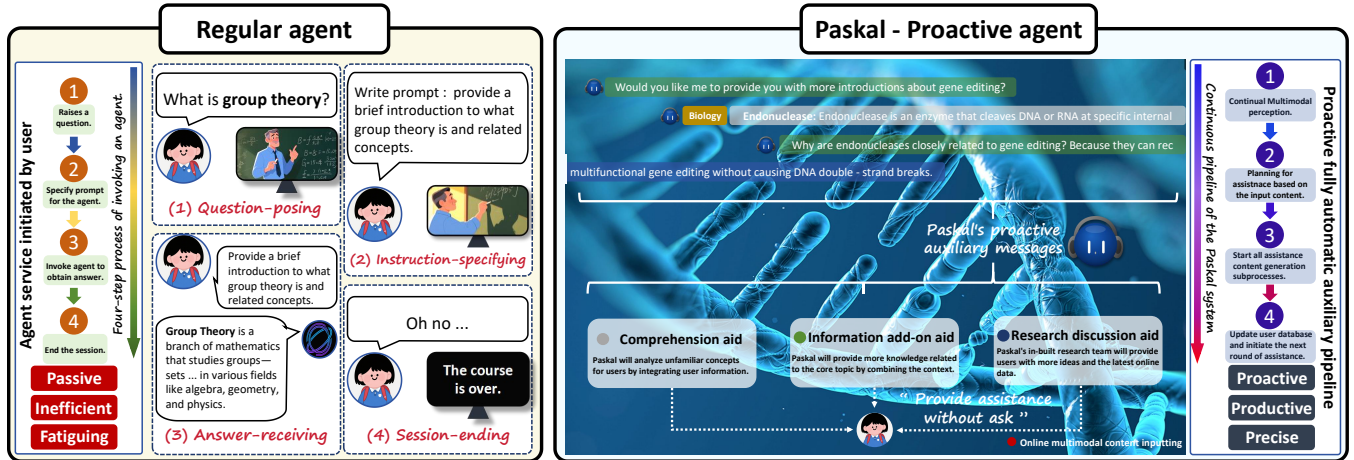


Figure 1: Comparison between passive AI assistants and our proactive agent, Pask. While conventional systems respond only to explicit prompts, following a "you ask, I answer" model, Pask shifts toward "answering before asking". It continuously monitors media, anticipates user needs, and provides context-aware explanations to support seamless understanding.

Abstract

We present Pask, a proactive AI agent that provides real-time, context-aware guidance and knowledge support in audio-centric media environments. Unlike passive assistants that follow the "you ask, I answer" model, Pask shifts toward "answering before asking" by continuously monitoring live audio, anticipating user needs, and

proactively offering conceptual explanations and semantic clarifications. It integrates three core components: a silent copilot for in-situ explanation, a structured knowledge base for factual grounding, and a private memory module for personalized adaptation. Pask enhances comprehension and communication in scenarios such as online learning, media consumption, and live meetings through sustained, intelligent guidance. A live demo is available at https://www.youtube.com/watch?v=ki_CKiV9Oyk.

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1 Introduction

“Like all communication systems, a learning system must be conversational.”

— Gordon Pask

In the era of large language and multimodal models [2–5, 8, 9], AI assistants have emerged as powerful digital butlers, capable of responding to diverse instructions and visual inputs. However, most remain fundamentally passive [1, 6, 7], relying on explicit user prompts to take action. This “*you ask, I answer*” paradigm assumes that users have the awareness and capacity to pause, formulate questions, and wait for responses. In real-world scenarios such as live lectures, video streams, or meetings, information flows rapidly and continuously, leaving little opportunity for active engagement. At such moments, conventional AI systems often remain silent, missing critical opportunities to support user understanding.

To address this limitation, we introduce the concept of a proactive AI agent. Rather than awaiting explicit user input, this system operates continuously, monitors dynamic media content, anticipates user needs, and delivers timely, context-aware guidance. It provides conceptual explanations, semantic clarifications, and concise summaries, supplemented by relevant knowledge, to facilitate real-time comprehension of complex material. By answering before being asked, this transition from reactive assistance to anticipatory instruction represents a crucial advancement toward AI systems that actively enhance human learning and communication.

We present **Pask**, an always-on AI copilot that embodies this transition. Operating through the user’s audio system, Pask integrates three core components: a silent copilot, a structured knowledge base, and a private memory module. The silent copilot monitors real-time audio and delivers transparent, non-intrusive explanations to guide users through unfamiliar content without disrupting their primary task. The knowledge base provides concise definitions, contextual information, and relevant suggestions aligned with the current media. The memory module maintains a personalized interaction history, enabling coherent and adaptive guidance over time. Pask supports use cases such as online learning, media consumption, and collaborative meetings, and aims to evolve into a proactive agent that is continuously engaged, knowledge-driven, and capable of delivering real-time, intelligent guidance.

2 Technology

This section outlines the system architecture and operational workflow of Pask, detailing its core components and how they collaboratively enable proactive, real-time guidance.

2.1 Front-end Perception and Database System.

Pask is a listening-based proactive AI agent operating in the audio modality, designed for scenarios where audio serves as the primary information channel, such as live lectures, video consumption, and meetings. It captures real-time system audio at a 16,000 Hz sampling rate via a front-end perception module, performing speech recognition, acoustic event detection, and linguistic analysis to extract contextual information for downstream AI assistance. User data is stored locally, systematically organized on a periodic basis, and continuously available for on-device access.

2.2 Proactive agents.

The Pask system integrates a suite of proactive agents that deliver real-time assistance based on continuous input from the audio stream. These agents provide support in three primary categories:

1. Hotword explanations: Pask identifies key concepts, figures, or terms within the recent context that are both semantically important and potentially unfamiliar to the user. Based on user background and dialogue history, it proactively provides concise explanations to support real-time understanding. **2. Open-information provision:** Pask connects the current context with relevant external or background knowledge, offering supplementary information to enrich user comprehension and promote broader, more open-ended exploration. **3. Novel ideas and suggestions:** Inspired by the AI scientist framework, Pask incorporates a lightweight research assistant agent that actively generates creative ideas, hypotheses, or decision-support insights grounded in the user’s ongoing discussion and the latest research developments.

2.3 Silent Assistance Display Mode

To ensure non-intrusive interaction, Pask employs a silent assistance display mode inspired by bullet comments, which unobtrusively scroll across the top of the screen. These comments are categorized into two types: (1) direct information display, which provides concise, context-aware explanations and knowledge snippets relevant to the ongoing media content; and (2) interactive prompts, which suggest optional extensions such as deeper dives, related facts, or task-specific tools. When users opt to engage with a prompt, detailed information is presented in a dedicated side panel or overlay. This lightweight, layered design enables Pask to deliver timely and proactive support while preserving the continuity and focus of the user’s primary activity.

3 User Experience

When users click on the Pask avatar, the system is activated and begins processing real-time computer audio input. Pask immediately initiates its overlay display by rendering subtitle-style assistance at the top of the screen. Each subtitle is annotated with a colored tag indicating its type (e.g., concept explanation, contextual hint, or interactive query), allowing users to distinguish information categories at a glance and access relevant content with minimal cognitive overhead. To minimize visual intrusion, all subtitles are semi-transparent and designed to blend naturally with the interface. Users retain lightweight control: a single click pauses or resumes subtitle scrolling, while a double-click expands the corresponding item—either presenting detailed elaboration (for informational content) or initiating active support (for interactive prompts). These interactions lead to an independent Pask interface, where users can review prior interactions, manage contextual history, or retrieve items from their personalized local knowledge base.

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